

An Earthquake Detection and Location Architecture for Continuous Data: Phase Picking, Association, Location, and Matched Filter (PALM)

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This supplemental content includes details in the PCA polarization analysis used in the PAL picker, the efficiency in each step of PALM, and figures that help validate the detection and location result in Ridgecrest.

PCA Polarization Filter

The polarization is calculated on three-channel waveforms with principal component analysis (PCA). The covariance matrix of a three-channel waveform at time t is defined as:

$$\sigma = \begin{bmatrix} \text{cov}(x, x) & \text{cov}(x, y) & \text{cov}(x, z) \\ \text{cov}(y, x) & \text{cov}(y, y) & \text{cov}(y, z) \\ \text{cov}(z, x) & \text{cov}(z, y) & \text{cov}(z, z) \end{bmatrix} \quad (1)$$

, where x, y, z is E, N, Z-channel data between t and $t+t_{pca}$, and t_{pca} is the window length for PCA calculation. We denote the eigenvector of σ as u_1, u_2, u_3 , and its eigenvalues as $\lambda_1, \lambda_2, \lambda_3$ ($\lambda_1 \geq \lambda_2 \geq \lambda_3$), respectively. Then, the polarization degree can be defined as:

$$r = 1 - \frac{\lambda_2 + \lambda_3}{2\lambda_1} \quad (2)$$

, and the principal direction of particle motion can be calculated as:

$$\cos \varphi(t) = |u_1(t) \cdot u_1(t_p)| \quad (3)$$

, where t_p is the arrival time of P wave. Finally, we can define the polarization filter for P wave as:

$$P_{filter}(t) = 1 - r(t) \cdot \cos \varphi(t) \quad (4)$$

Our design of polarization filter only processes the high-polarized part of P wave, and avoids the waveform with low polarization degree, where multi-phases may exist. This is different from that in (Ross and Ben-Zion, 2014), and can avoid the picking deviation caused by falsely suppressed S wave.

Efficiency of PALM

The efficiency of PALM is tested on a workstation computer, equipped with Intel Xeon CPU E5-2620 (16 physical cores, 2.10GHz), and two GTX 1080Ti GPUs. Data are stored on one 8-T Western Digital hard drive (WD82EJRX).

PAL is run with multiprocessing. For a single process, the PAL picker takes ~1-hour to pick ~50,000 P&S pairs on 2019-07-08; the PAL associator associate these picks within 2-hours. The total running time in this study is about 5-hours, with one process for one day's detection. The total location process takes less than 30-min to accomplish.

MESS is implemented with GPU support. In this study, we down-sampled the data to 50Hz, for fast calculation of cross-correlation. One template of 15 station records takes about 1.5s to detect one day's data. The total running time for MESS is ~50-hurs

in this study. The association and relocation of MESS detections in this study is ~20min.

Usage Guide of PALM

PALM, as a software, is based on Python, Obspy, Numpy, Scipy, and Pytorch. It is a set of codes that runs on certain environment, which is made easy by Anaconda: with the `.yaml` file provided in the Github link of PAL & MESS (See Data and Resources Section), the proper environment can be configured with one single command. Example data is also provided in the Github repository.

The parameters in PAL is divided in to picker and associator, whose function are explained in the “PAL: Picking, Association, and Location” Section. The picker parameters are stable under most cases, while only s_win (Δt_s in Equ. 4) should be adjusted based on the inter-station distance. A rule of thumb is that a circle with radius $10*s_win$ should cover at least four stations in your objective region. The default parameters of PAL associator also works for general cases. The ot_dev (maximum origin time deviation, Equ. 7) and max_res (maximum P residual, Equ. 8) can be larger if your network is sparse. A rule of thumb is to estimate a maximum uncertainty of the local wave speed, and see how large the origin time and P residual can be mis-estimated. The min_sta (minimum number of station to declare an event detection) should always be no less than 3, and can be larger if you have a rather dense network. Note that PAL works for certain number of stations, which range from about 5 to 100 empirically, and depend on the noise level of the stations. For even larger network, we suggest users to divide it into sub-networks to get better association performance.

MESS provide both CPU and GPU version. Except for clusters that contain hundreds of CPU cores, GPU version would provide much high efficiency. To get the best MESS performance, the window length should vary by your network configuration: The $temp_win_det$ (length of detection window, i.e. Δt_{temp} in Equ. 10) should cover both P&S phases, the $temp_win_p$ and $temp_win_s$ (the phase picking window for P & S phases) should be short enough to avoid overlapping, but should be at least 1s to include enough number of wiggles. The $expand_len$ (length of CC peak expansion, i.e. Δt_{exp} in Equ. 11) controls the searching radius away from the template location, e.g. 2s correspond to ~6km assuming 6km/s P wave speed. The $trig_thres$ (minimum CC value to declare a detection) controls the detectability: a high threshold gives less detections but stronger links between templates and detections, which provide robust relocation; a low threshold provide larger detection number, but may also cause more poorly located or false alarmed events. Empirically, a threshold between 0.25 and 0.35 would provide a good baseline result.

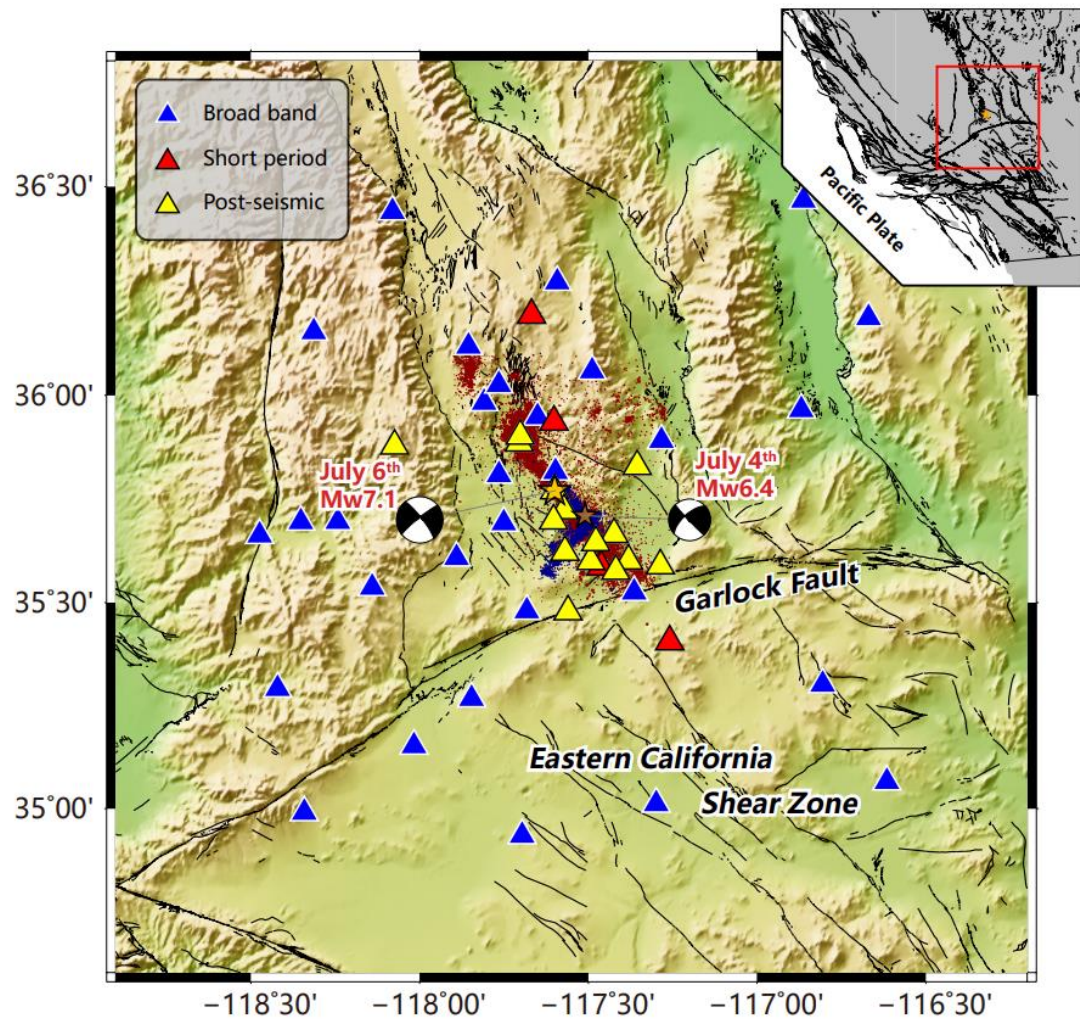


Figure S1. Station distribution and tectonic setting. Blue, red and yellow triangles denote broadband, short period and post-seismic stations. Foreshock and mainshock sequences detected by SCSN are plotted in dark blue and dark red dots.

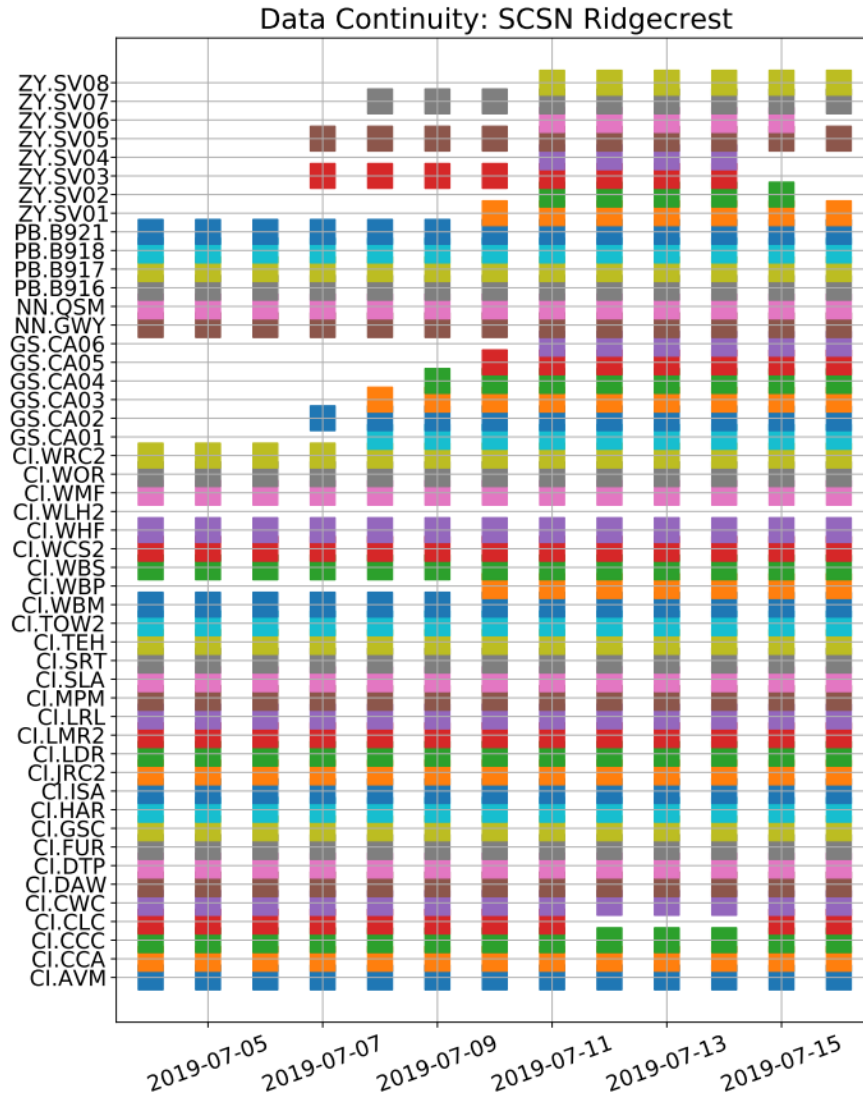


Figure S2. Continuity of SCSN data. Days, when data are available from SCSN, are plotted with square markers.

PAL Picker on 20190706210524.59 - CI.CLC

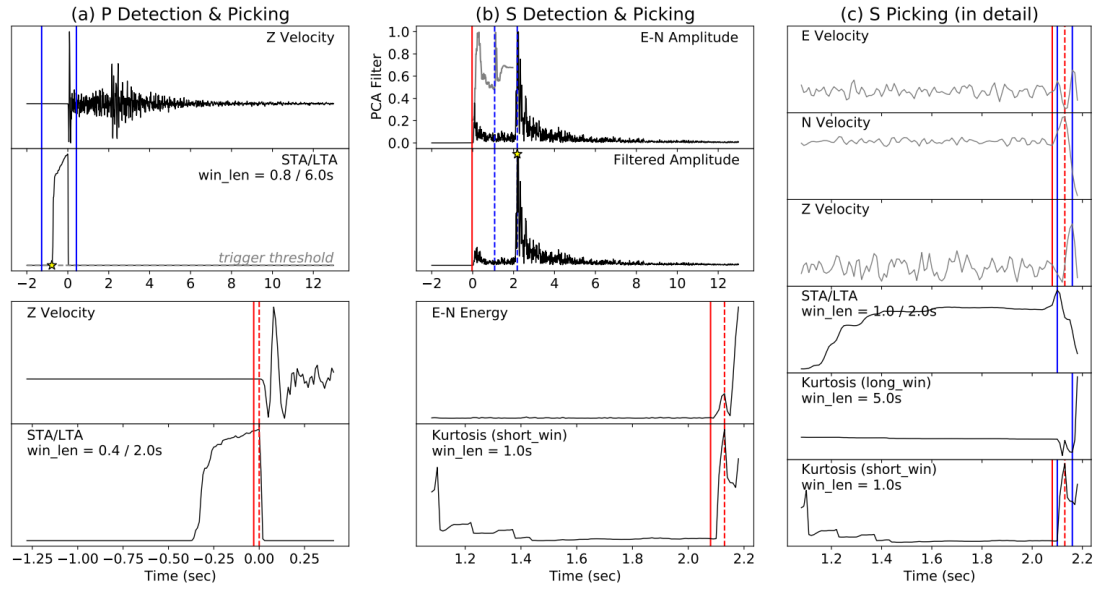


Figure S3. Example PAL pick for a high-SNR record. Phase windows are bounded by blue vertical lines. Dashed and solid red lines plot initial and final (corrected) phase picks.

PAL Picker on 20190704210030.01 - PB.B916

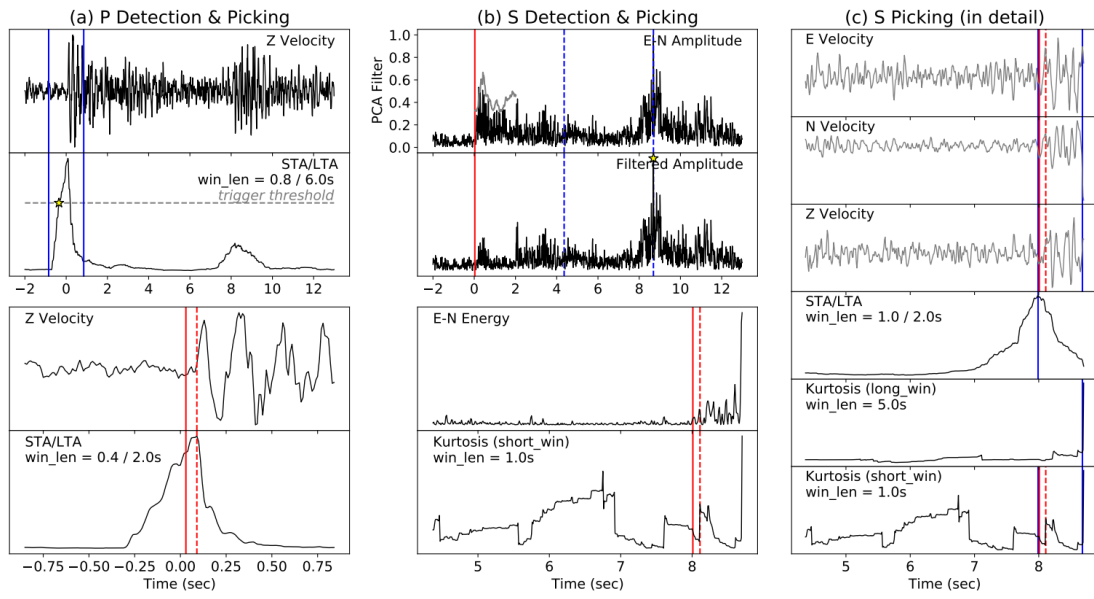


Figure S4. Example pick for a typical earthquake record. Markers have the same meaning as Figure S3.

PAL Picker on 20190704174254.13 - CI.DAW

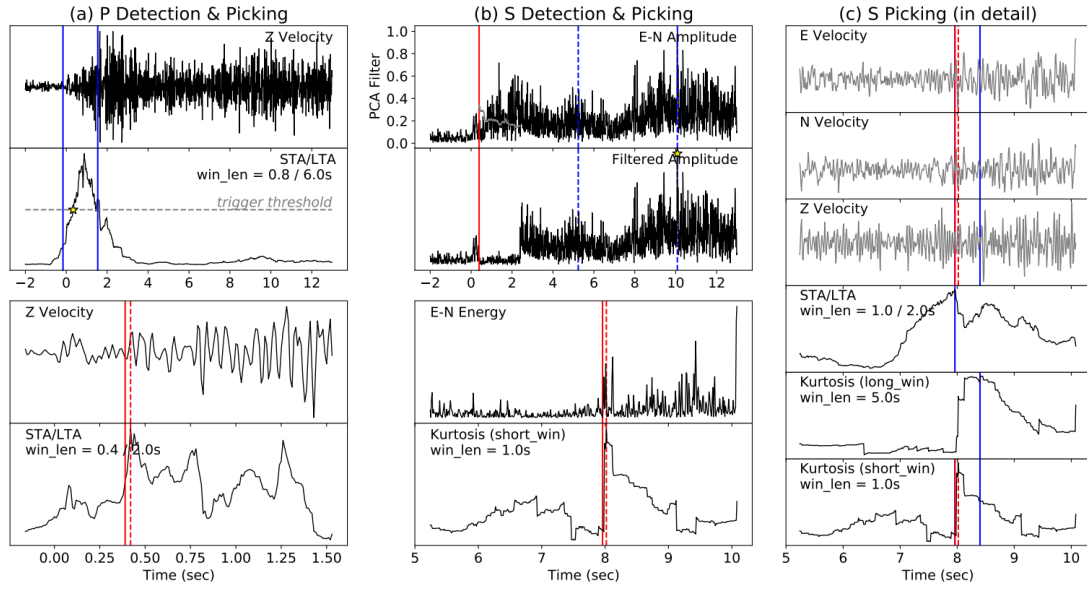


Figure S5. Example pick for a low-SNR waveform. Markers have the same meaning as Figure S3.

PAL Picker on 20190704161343.37 - CI.DAW

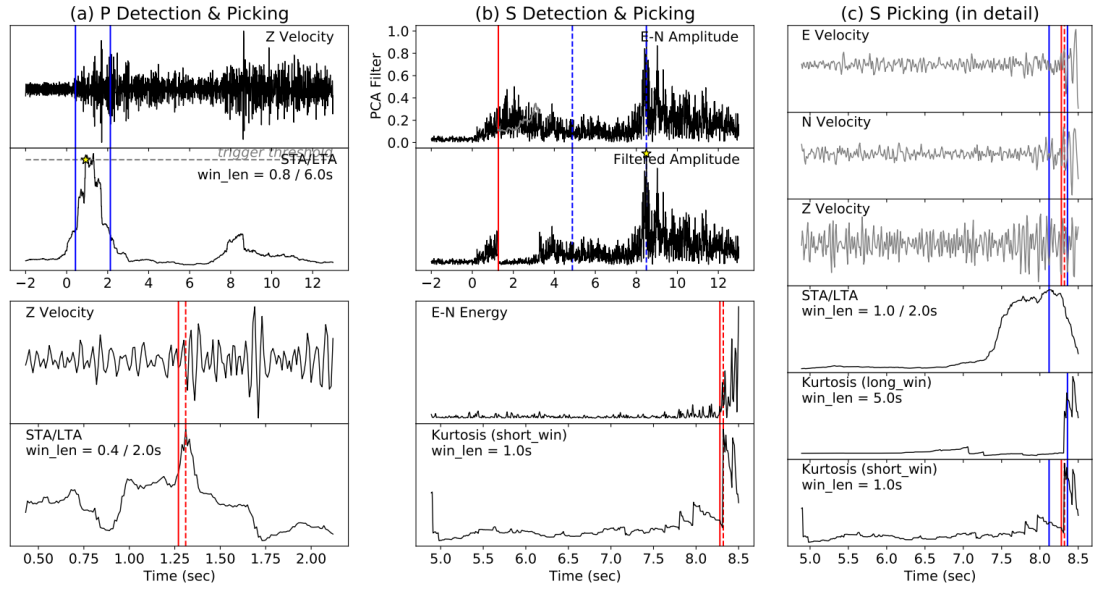


Figure S6. Example false P pick by PAL. Markers have the same meaning as Figure S3.

PAL Picker on 20190704194904.98 - CI.WBS

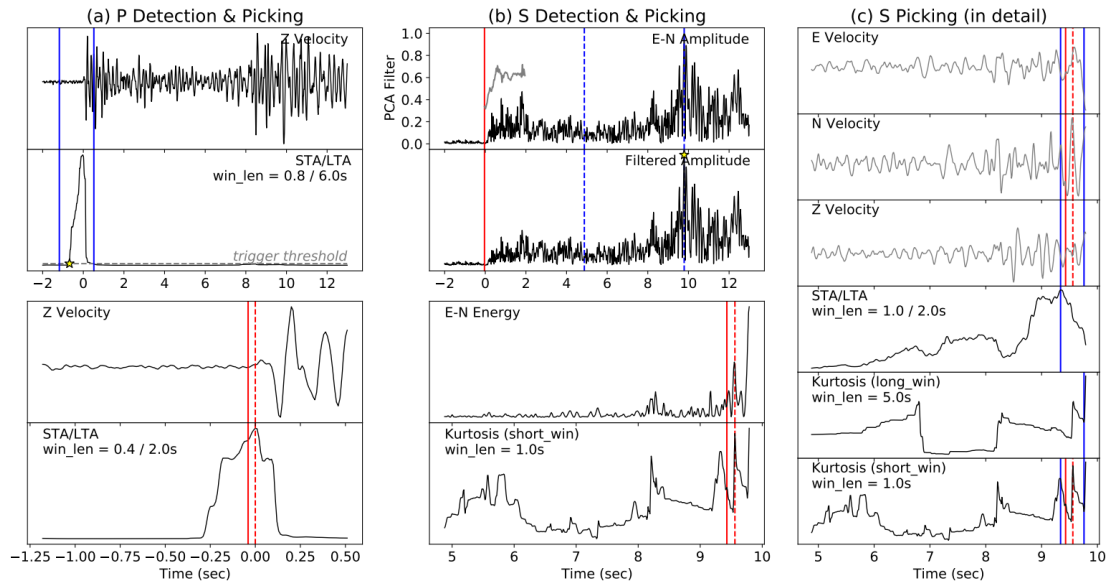


Figure S7. Example false S pick by PAL. Markers have the same meaning as Figure S3.

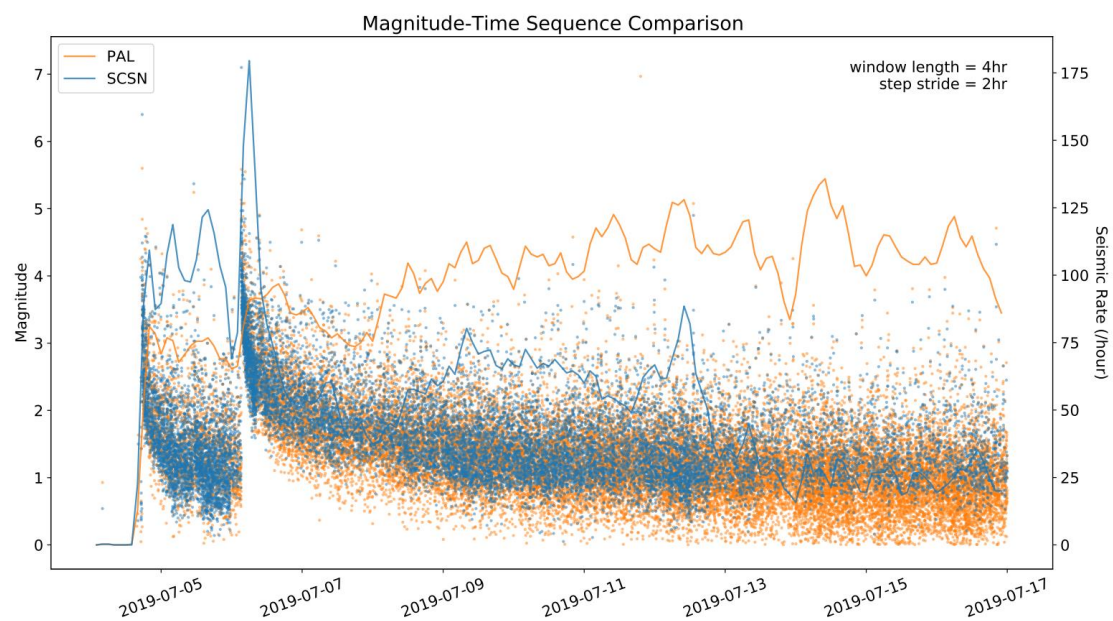


Figure S8. Magnitude-time sequence comparison for PAL and SCSN catalog. Orange and blue represent PAL and SCSN catalog, respectively. Earthquakes and seismic rates are plotted in dots and lines, respectively.

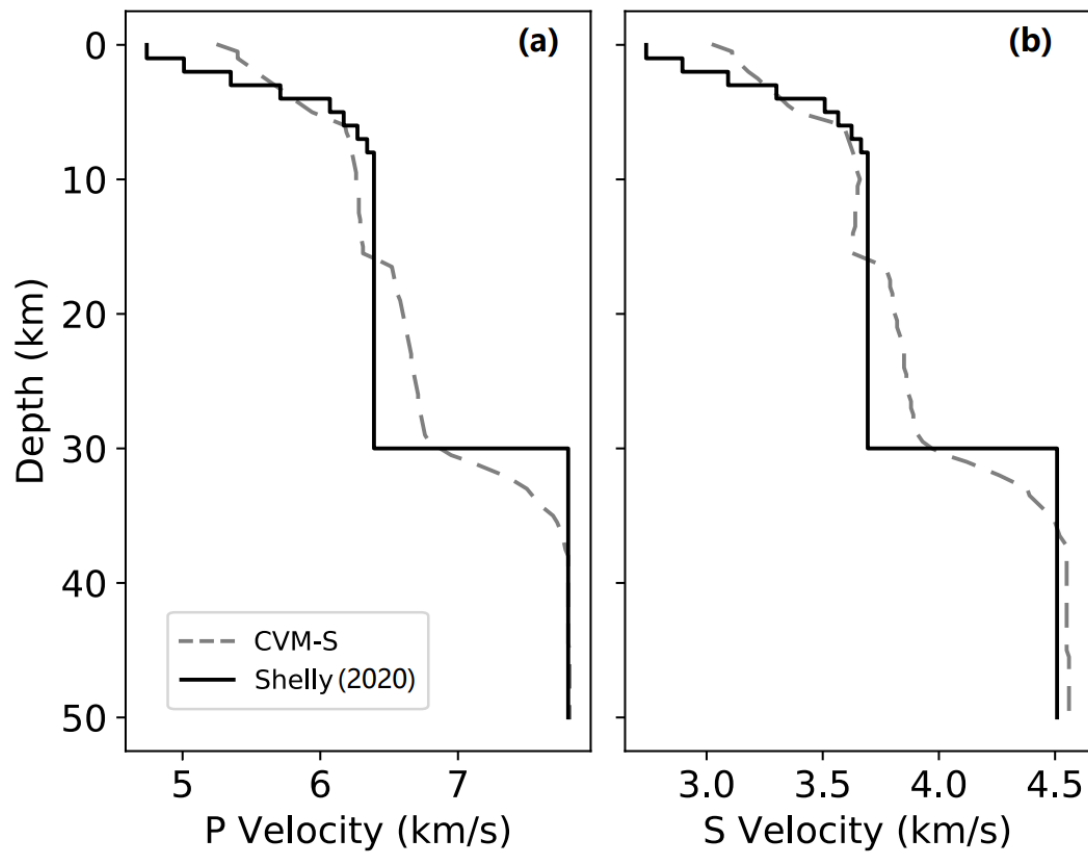


Figure S9. Velocity model comparison. The solid black line and dashed gray line plot Shelly (2020) mode and CVM-S model, respectively.

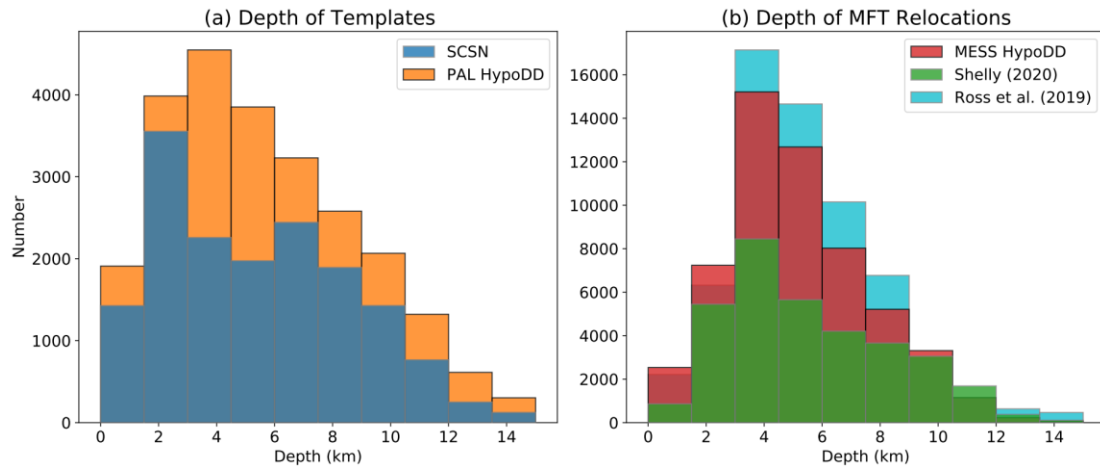


Figure S10. Comparison of depth distribution. (a) and (b) plot depth distribution for templates and MFT relocation results, respectively.

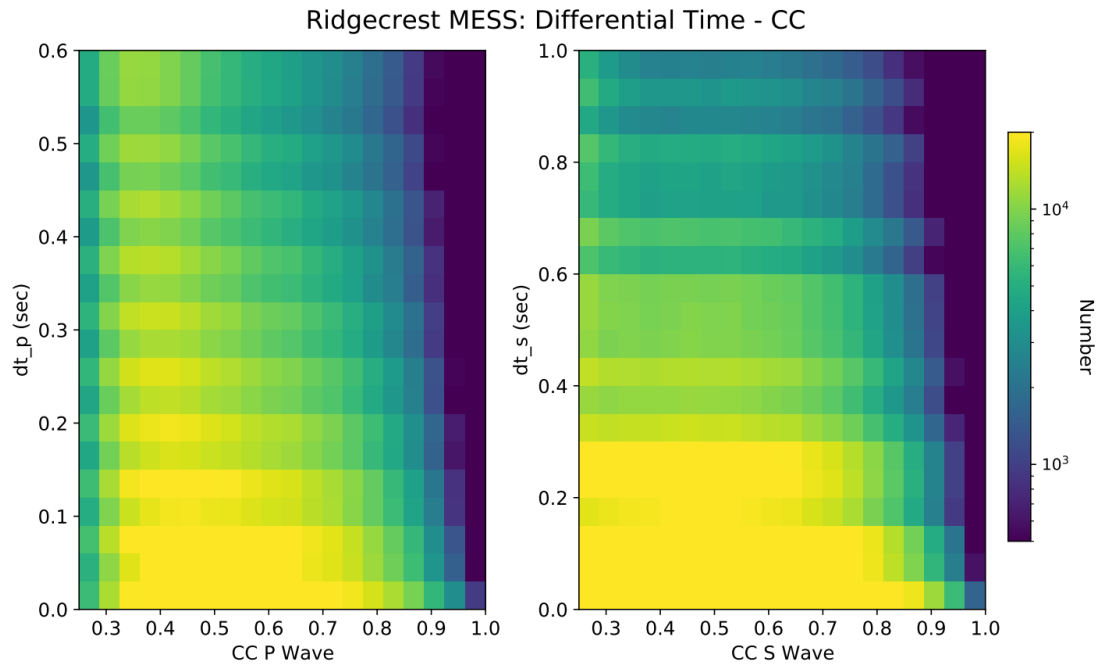


Figure S11. Distribution of differential travel time and CC values. The number of dt measurements in each grid is represented by color.

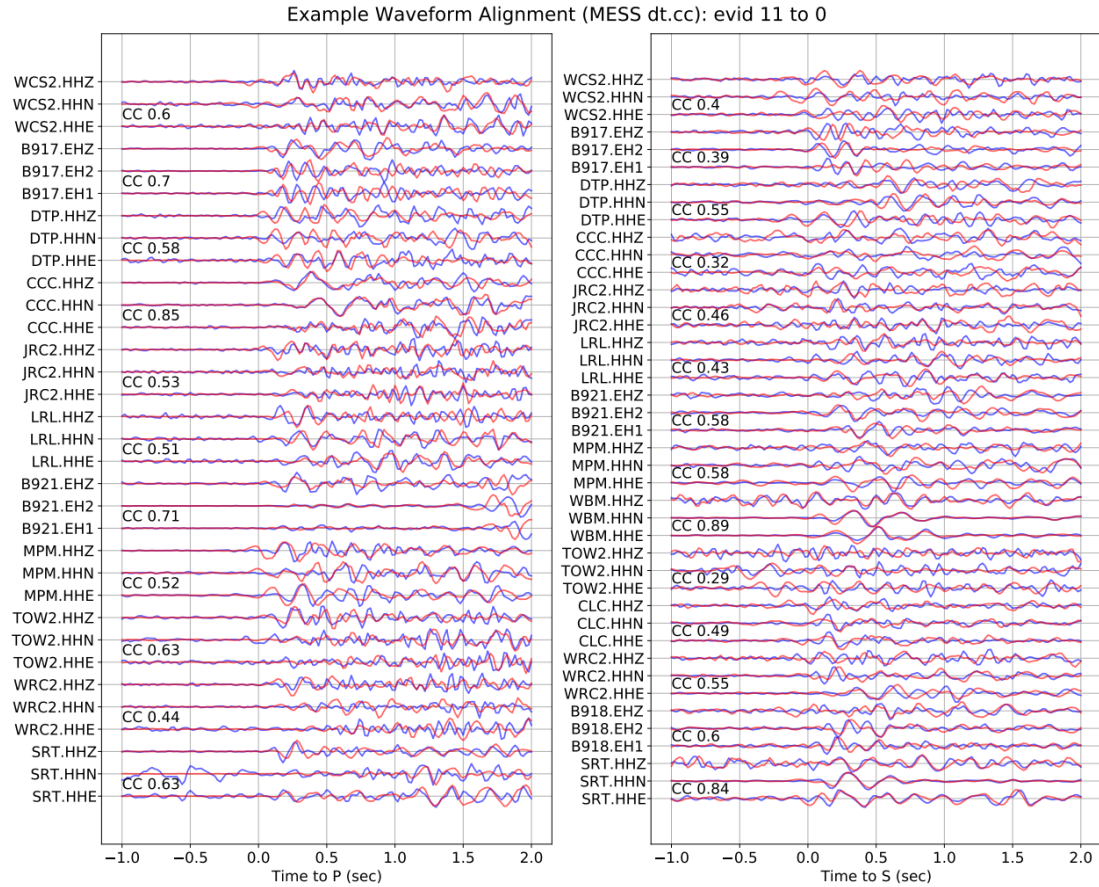


Figure S12. Example waveform alignment by MESS determined differential travel time. Blue and red lines plot the waveforms of event 11 and 0, respectively.

References

Ross, Z. E., and Y. Ben-Zion (2014), Automatic picking of direct P, S seismic phases and fault zone head waves, *Geophysical Journal International*, 199(1), 368-381, doi:10.1093/gji/ggu267.